Formation of PDEs: The Idea

- Conversely, a family of surfaces may be described by a PDE.
- A PDE can be formed by eliminating.
- (i) arbitrary constants from a relation, or.
- (ii) arbitrary functions from a relation.
- Compare ODEs: $y = c\,x^2$ eliminating c gives $x\,y' = 2y$.
- For surfaces $u = u(x, y)$ we differentiate partially w.r.t. x and y.
- Then eliminate the constants / functions from the resulting equations.
- This helps us understand what the general solution of a PDE looks like.



Eliminating Arbitrary Constants

- Let $u = u(x, y)$, $p = u_x$, $q = u_y$, and consider a two-parameter family

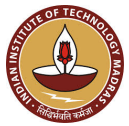
$$f(x, y, u, a, b) = 0, \qquad a, b \in \mathbb{R}. \tag{1}$$

- Differentiate (1) partially w.r.t. x and y (chain rule):

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial u}\, p = 0 \tag{2}$$

$$\frac{\partial f}{\partial y} + \frac{\partial f}{\partial u}\, q = 0. \tag{3}$$

- Eliminate a and b from (1)–(3).
- Result: a first order PDE $F(x, y, u, p, q) = 0$.



Number of Constants and the Order

- Let k = number of arbitrary constants, n = number of independent variables.
- If $k = n$: elimination gives a PDE of first order.
- If $k > n$: we need higher derivatives, so the PDE is of second or higher order.
- If $k < n$: elimination is possible in more than one way; we may get several first order PDEs.
- For $n = 2$ and $k = 2$: three equations (1), (2), (3) and two unknowns a, b.
- Exactly enough to eliminate a, b and get one relation among x, y, u, p, q.
- Usually the result is a nonlinear PDE.



Example: $u = (x^2 + a)(y^2 + b)$

- Two constants a, b, two variables $x, y \Rightarrow$ first order PDE.
- $p = u_x = 2x\,(y^2 + b)$.
- $q = u_y = 2y\,(x^2 + a)$.
- Multiply: $p\,q = 4xy\,(x^2 + a)(y^2 + b)$.
- Use the given relation to replace the product:

$$p\,q = 4xy\,u.$$

- This is a first order nonlinear PDE.
- Each surface of the family satisfies it.



Example: Spheres with Centre on the u-axis

- Find the PDE of all spheres whose centres lie on the u-axis (vertical axis).
- Family: $x^2 + y^2 + (u - a)^2 = b^2, \quad a, b \in \mathbb{R}$, with $u = u(x, y)$.
- Differentiate w.r.t. x: $2x + 2(u - a)\, p = 0$.
- Differentiate w.r.t. y: $2y + 2(u - a)\, q = 0 \Rightarrow u - a = -\dfrac{y}{q}$.
- Substitute in the first: $x - \dfrac{y}{q}\, p = 0$.
- Hence

$$y\, p - x\, q = 0.$$

- Note: the result is linear, although the family is nonlinear in a.



Example: Planes $u = ax + by + ab$

- $p = a, \quad q = b.$
- Substitute back for a and b:

$$u = p\,x + q\,y + p\,q.$$

- A first order nonlinear PDE of Clairaut type.
- Each plane of this two-parameter family is a solution.
- We will see that this PDE has another solution not in the family.
- Similarly $u = ax + by + a^2 + b^2$ gives $u = px + qy + p^2 + q^2$.
- In general $u = ax + by + g(a, b)$ gives $u = px + qy + g(p, q)$.

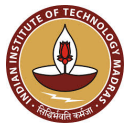


Example: Unit Spheres with Centres in the xy-plane

- Family: $(x-a)^2+(y-b)^2+u^2=1$.
- W.r.t. x: $(x-a)+u\,p=0 \Rightarrow x-a=-u\,p$.
- W.r.t. y: $(y-b)+u\,q=0 \Rightarrow y-b=-u\,q$.
- Substitute in the family:

$$u^2p^2+u^2q^2+u^2=1.$$

- Required PDE: $u^2\left(p^2+q^2+1\right)=1$.
- A nonlinear first order PDE.
- The planes $u=\pm 1$ also satisfy it (they touch every sphere).



Practice: Eliminating Constants

- Form the PDE by eliminating a, b (and c) from:
- (i) $u = (x-a)^2 + (y-b)^2$ Ans: $p^2 + q^2 = 4u$.
- (ii) $u = ax^2 + bxy + cy^2$ (three constants $\Rightarrow$ second order).
- (iii) $u = (ax + y)^2 + b$.
- (iv) $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} + \dfrac{u^2}{c^2} = 1$.
- (v) $u = ax + by + \sqrt{a^2 + b^2}$.
- (vi) $u = ae^{bx}\sin(by)$.
- Before eliminating, predict the order of the resulting PDE.



Eliminating an Arbitrary Function

- Let $\varphi = \varphi(x, y, u)$, $\psi = \psi(x, y, u)$ be known functions, $u = u(x, y)$.
- Consider the relation with an arbitrary function G:

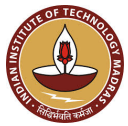
$$G(\varphi, \psi) = 0. \tag{1}$$

- Equivalent forms: $\varphi = g(\psi)$ or $\psi = h(\varphi)$, with g, h arbitrary.
- Differentiate (1) partially w.r.t. x and y (chain rule):

$$G_\varphi \left(\varphi_x + \varphi_u \, p\right) + G_\psi \left(\psi_x + \psi_u \, p\right) = 0 \tag{2}$$

$$G_\varphi \left(\varphi_y + \varphi_u \, q\right) + G_\psi \left(\psi_y + \psi_u \, q\right) = 0. \tag{3}$$

- Assume G_φ and G_ψ are not both zero.



Eliminating G_φ and G_ψ

- (2), (3) form a homogeneous linear system in (G_φ, G_ψ).
- A non-trivial solution exists only if the determinant vanishes:

$$\begin{vmatrix} \varphi_x + \varphi_u p & \psi_x + \psi_u p \\ \varphi_y + \varphi_u q & \psi_y + \psi_u q \end{vmatrix} = 0.$$

- Expanding:

$$(\varphi_x + p\varphi_u)(\psi_y + q\psi_u) - (\varphi_y + q\varphi_u)(\psi_x + p\psi_u) = 0.$$

- The pq terms cancel: $pq(\varphi_u\psi_u - \varphi_u\psi_u) = 0$.
- So the resulting PDE is linear in p and q.



Result: A Quasi-linear PDE

- Collecting terms we obtain

$$(\varphi_y\psi_u - \varphi_u\psi_y)\,p + (\varphi_u\psi_x - \varphi_x\psi_u)\,q = \varphi_x\psi_y - \varphi_y\psi_x.$$

- i.e. $a(x,y,u)\,p + b(x,y,u)\,q = c(x,y,u)$, where (Jacobians)

$$a = \frac{\partial(\varphi,\psi)}{\partial(y,u)}, \quad b = \frac{\partial(\varphi,\psi)}{\partial(u,x)}, \quad c = \frac{\partial(\varphi,\psi)}{\partial(x,y)}.$$

- Eliminating one arbitrary function always gives a quasi-linear first order PDE.
- This is exactly Lagrange's equation; its solution will reverse this process.



Example: $u = x^n f(y/x)$

- Equivalently $G\big(u/x^n,\ y/x\big) = 0$; direct differentiation is quicker.
- $p = nx^{n-1} f(y/x) + x^n f'(y/x) \cdot \Big(-\dfrac{y}{x^2} \Big)$.
- $q = x^n f'(y/x) \cdot \dfrac{1}{x} \ \Rightarrow\ f'(y/x) = \dfrac{q}{x^{n-1}}$.
- Substitute: $p = nx^{n-1} f - x^{n-2}\, y\, \dfrac{q}{x^{n-1}}$.
- Multiply by x: $x\,p = nx^n f - y\,q$.
- Hence

$$x\,p + y\,q = n\,u \qquad \text{(linear)}.$$

- This is Euler's relation for homogeneous functions of degree n.



Example: $u = e^{mx} f(x+y)$

- Equivalently $G\left(u\, e^{-mx},\ x+y\right) = 0$.
- By the chain rule:
- $p = m\, e^{mx} f(x+y) + e^{mx} f'(x+y) \cdot 1$.
- $q = e^{mx} f'(x+y) \cdot 1$.
- Subtract to eliminate f':

$$p - q = m\, e^{mx} f(x+y) = m\, u.$$

- Required PDE: $p - q = m\, u$ (linear).
- Remember to use the chain rule on the argument of f.



Example: $G(xy + u^2,\ x + y + u) = 0$

- Let $\varphi = xy + u^2$, $\psi = x + y + u$.
- Differentiate $G(\varphi, \psi) = 0$ w.r.t. x:

$$G_\varphi\,(y + 2u\,p) + G_\psi\,(1 + p) = 0.$$

- Differentiate w.r.t. y:

$$G_\varphi\,(x + 2u\,q) + G_\psi\,(1 + q) = 0.$$

- Eliminate G_φ, G_ψ:

$$\begin{vmatrix} y + 2up & 1 + p \\ x + 2uq & 1 + q \end{vmatrix} = 0.$$



Example (contd.)

- Expanding: $(y + 2up)(1 + q) - (x + 2uq)(1 + p) = 0$.
- $y + yq + 2up + 2upq - x - xp - 2uq - 2upq = 0$.
- The upq terms cancel.
- Collecting:
$$(2u - x)\, p + (y - 2u)\, q = x - y.$$
- This is a quasi-linear PDE (coefficients contain u).
- Check with the Jacobian formula: $a = \varphi_y \psi_u - \varphi_u \psi_y = x - 2u$, etc. (overall sign immaterial).



Example: $xyu = f(x + y + u)$

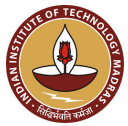
- Let $\varphi = xyu$, $\psi = x + y + u$; then $G(\varphi, \psi) = 0$.
- W.r.t. x: $G_\varphi\,(yu + xy\,p) + G_\psi\,(1 + p) = 0$.
- W.r.t. y: $G_\varphi\,(xu + xy\,q) + G_\psi\,(1 + q) = 0$.
- Eliminating:

$$\begin{vmatrix} yu + xyp & 1 + p \\ xu + xyq & 1 + q \end{vmatrix} = 0.$$

- $(yu + xyp)(1 + q) - (xu + xyq)(1 + p) = 0$.
- The $xypq$ terms cancel, leaving

$$x(y - u)\,p + y(u - x)\,q = u(x - y).$$

- A quasi-linear PDE.



More Examples with Arbitrary Functions

- $u = f(x^2 + y^2)$: $p = 2xf'$, $q = 2yf' \Rightarrow y\,p - x\,q = 0$.
- Compare with the spheres example: same PDE (surfaces of revolution about the u-axis).
- $u = f(x - ct)$ (with $y = t$): $u_x = f'$, $u_t = -cf' \Rightarrow$ $u_t + c\,u_x = 0$.
- This is the transport equation: profiles move with speed c.
- $u = f(x + ct) + g(x - ct)$: two arbitrary functions.
- One needs second derivatives to eliminate both.
- Result: $u_{tt} = c^2 u_{xx}$, the wave equation (second order).
- Rule: k arbitrary functions usually lead to a PDE of order k.



Practice: Eliminating Functions

- Form PDEs by eliminating the arbitrary function(s):
- (i) $u = f(x^2 - y^2)$ Ans: $y\,p + x\,q = 0$.
- (ii) $u = f\left(\frac{y}{x}\right)$ Ans: $x\,p + y\,q = 0$.
- (iii) $u = y^2 + 2f\left(\frac{1}{x} + \ln y\right)$.
- (iv) $G(x + y + u,\ x^2 + y^2 - u^2) = 0$.
- (v) $u = f(x + iy) + g(x - iy)$ (second order).
- (vi) $G\left(\frac{y}{u},\ x^2 + y^2\right) = 0$.
- Identify whether each result is linear or quasi-linear.



Complete Integral

- Consider $F(x, y, u, p, q) = 0$.
- A two-parameter family $f(x, y, u, a, b) = 0$ satisfying the PDE for all a, b is called a complete solution, complete integral or primitive.
- The parameters must be genuinely independent.
- i.e. the matrix $\begin{pmatrix} u_a & p_a & q_a \\ u_b & p_b & q_b \end{pmatrix}$ has rank 2.
- Example: $u = (x^2 + a)(y^2 + b)$ is a complete integral of $pq = 4xyu$.
- Example: $u = ax + by + ab$ is a complete integral of $u = px + qy + pq$.
- The number of parameters equals the number of independent variables.



General Integral

- From a complete integral $u = \Phi(x, y, a, b)$, put $b = h(a)$ with h arbitrary.
- We get a one-parameter family $u = \Phi(x, y, a, h(a))$.
- Its envelope is found by eliminating a from

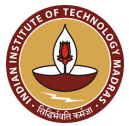
$$u = \Phi(x, y, a, h(a)), \qquad \Phi_a + \Phi_b\, h'(a) = 0.$$

- The envelope is again a solution (it touches each member).
- Since h is arbitrary, this gives a solution with an arbitrary function: the general integral.
- For quasi-linear PDEs: general solution $G(\varphi, \psi) = 0$ with G arbitrary.
- Here $\varphi = c_1$ and $\psi = c_2$ come from Lagrange's method.



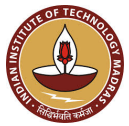
Singular Integral

- The envelope of the full two-parameter family, if it exists, is the singular integral.
- Eliminate a, b from $f = 0$, $f_a = 0$, $f_b = 0$.
- Example: $u = px + qy + pq$ with complete integral $u = ax + by + ab$.
- $\partial/\partial a$: $0 = x + b$; $\partial/\partial b$: $0 = y + a$.
- So $a = -y$, $b = -x$ and $u = -xy - xy + xy$.
- Singular integral: $u = -xy$.
- Check: $p = -y$, $q = -x$: $px + qy + pq = -xy - xy + xy = -xy$ ✓.
- It is not obtained from the complete integral by any choice of a, b.



Particular Solution and Side Conditions

- A particular solution is obtained by specific choices of constants or of the arbitrary function.
- Example: $u = f(x^2 + y^2)$ solves $y\,p - x\,q = 0$.
- Particular solutions: $u = x^2 + y^2$, $u = e^{-(x^2+y^2)}$, $u = 5$.
- To pick one solution we impose extra data, e.g. the values of u on a curve.
- For $u(x, y)$: data $u(x, y_0) = g(x)$ or $u(x_0, y) = g(y)$.
- More generally: u prescribed on a curve in the xy-plane.
- Geometrically: the solution surface must contain a given space curve.
- This leads to the Cauchy problem, studied later.



Integral Surfaces

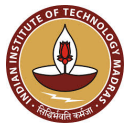
- Consider the quasi-linear first order PDE

$$a(x, y, u)\, u_x + b(x, y, u)\, u_y = c(x, y, u). \qquad (*)$$

- A solution $u = u(x, y)$ represents a surface S in (x, y, u)-space.
- Such a surface is called an integral surface of $(*)$.
- Example: the sphere $x^2 + y^2 + (u - c)^2 = k^2$ is an integral surface of

$$y\, u_x - x\, u_y = 0.$$

- Similarly the paraboloid $u = x^2 + y^2$ and all surfaces of revolution about the u-axis.
- Question: what geometric property characterises integral surfaces of $(*)$?



Normal to the Surface

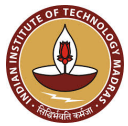
- Write the surface $u = u(x, y)$ as $\Phi(x, y, u) := u(x, y) - u = 0$.
- From calculus, $\nabla\Phi$ is normal to the level surface.
- Hence the normal is

$$\vec{n} = (u_x,\ u_y,\ -1) = (p,\ q,\ -1).$$

- The direction cosines of the normal are proportional to $(p, q, -1)$.
- Now $(*)$ reads

$$(a,\ b,\ c) \cdot (p,\ q,\ -1) = a\,p + b\,q - c = 0.$$

- So the vector (a, b, c) is perpendicular to the normal $\vec{n}$.



Tangency Condition

- At each point (x, y, u) the PDE assigns the vector $\vec{V} = (a, b, c)$.
- $\vec{V}$ defines a direction field in (x, y, u)-space.
- Since $\vec{V} \perp \vec{n}$, the vector $\vec{V}$ lies in the tangent plane of S.
- Thus any integral surface S is tangent to the field (a, b, c) at every point.
- Conversely, if a smooth surface is tangent to $\vec{V}$ everywhere, then $ap + bq = c$ on it.
- So: *integral surfaces = surfaces tangent to the direction field* (a, b, c).
- This reduces the PDE to the study of curves along the field.



Derivation of the Auxiliary Equations

- Along any curve on S: $du = u_x\,dx + u_y\,dy$, i.e.

$$p\,dx + q\,dy - du = 0. \tag{1}$$

- The PDE gives

$$a\,p + b\,q - c = 0. \tag{2}$$

- Choose curves on S whose tangent (dx, dy, du) points along (a, b, c).
- Comparing (1) and (2), (a, b, c) and (dx, dy, du) are proportional.
- This yields Lagrange's auxiliary (characteristic) equations

$$\frac{dx}{a(x,y,u)} = \frac{dy}{b(x,y,u)} = \frac{du}{c(x,y,u)}\ (= dt).$$

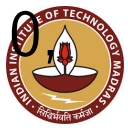


Characteristic Curves

- The auxiliary equations form a system of ODEs:

$$\frac{dx}{dt} = a, \qquad \frac{dy}{dt} = b, \qquad \frac{du}{dt} = c.$$

- Its solution curves are called characteristic curves.
- Their projections on the xy-plane are the characteristic base curves.
- Through each point passes exactly one characteristic (when a, b, c are smooth).
- The system has two independent first integrals $\varphi(x, y, u) = c_1$, $\psi(x, y, u) = c_2$.
- Each characteristic is an intersection of two surfaces $\varphi = c_1$, $\psi = c_2$.
- Key fact: an integral surface is a union of characteristic curves.
- Picking a one-parameter family of characteristics, $G(c_1, c_2) = 0$, gives a surface.



Intersection of Two Integral Surfaces

- Proposition. If two integral surfaces S_1, S_2 intersect transversally along a curve Γ, then Γ is a characteristic curve.
- Let $P \in \Gamma$. Both tangent planes at P contain the direction (a, b, c).
- Two distinct planes meet in exactly one line.
- So their line of intersection is along (a, b, c).
- But the tangent to Γ at P lies in both tangent planes.
- Hence the tangent to Γ is parallel to (a, b, c) at every point.
- Therefore Γ satisfies $\frac{dx}{a} = \frac{dy}{b} = \frac{du}{c}$, i.e. it is a characteristic.
- Consequence: two different solutions can only branch apart along characteristics.



Lagrange's Method: Statement

- No general method exists for fully nonlinear first order PDEs.
- For quasi-linear PDEs, Lagrange's method reduces the problem to solving ODEs.
- Theorem. Let a, b, c be continuous (with continuous first derivatives) in a domain and not all zero at any point.
- Let $\varphi(x, y, u) = c_1$, $\psi(x, y, u) = c_2$ be two independent solutions of

$$\frac{dx}{a} = \frac{dy}{b} = \frac{du}{c}.$$

- Then the general solution of $a\,p + b\,q = c$ is

$$G(\varphi, \psi) = 0.$$

- where G is an arbitrary (differentiable) function of two variables.



Proof: Step 1

- Since $\varphi = c_1$ along the characteristics, $d\varphi = \varphi_x dx + \varphi_y dy + \varphi_u du = 0$.
- Along them $(dx, dy, du) \propto (a, b, c)$, hence

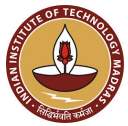
$$a\,\varphi_x + b\,\varphi_y + c\,\varphi_u = 0.$$

- Similarly

$$a\,\psi_x + b\,\psi_y + c\,\psi_u = 0.$$

- Solve these two linear equations for the ratios $a : b : c$ (cross-multiplication):

$$\frac{a}{\varphi_y\psi_u - \varphi_u\psi_y} = \frac{b}{\varphi_u\psi_x - \varphi_x\psi_u} = \frac{c}{\varphi_x\psi_y - \varphi_y\psi_x}.$$



Proof: Step 2

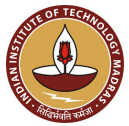
- In Jacobian notation:

$$\frac{a}{\partial(\varphi,\psi)/\partial(y,u)} = \frac{b}{\partial(\varphi,\psi)/\partial(u,x)} = \frac{c}{\partial(\varphi,\psi)/\partial(x,y)}.$$

- Now let $u(x,y)$ satisfy $G(\varphi,\psi)=0$ for some differentiable G.
- From the formation of PDEs (eliminating G) we know u satisfies

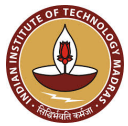
$$\frac{\partial(\varphi,\psi)}{\partial(y,u)}\,p + \frac{\partial(\varphi,\psi)}{\partial(u,x)}\,q = \frac{\partial(\varphi,\psi)}{\partial(x,y)}.$$

- The coefficients here are proportional to a, b, c.
- Dividing by the common factor: $a\,p + b\,q = c$.



Proof: Step 3 and Conclusion

- Hence every surface $G(\varphi, \psi) = 0$ is an integral surface.
- It contains one arbitrary function, as expected for a first order PDE.
- Conversely, every solution can be shown to be of this form (locally).
- Indeed, on any solution surface, φ and ψ are functionally dependent.
- i.e. $\partial(\varphi, \psi)/\partial(x, y) = 0$ along $u(x, y)$, giving a relation $G(\varphi, \psi) = 0$.
- Therefore $G(\varphi, \psi) = 0$ is the general solution.
- Equivalent forms: $\varphi = g(\psi)$ or $\psi = h(\varphi)$, g, h arbitrary functions of one variable.



Working Procedure

- Step 1: Write the PDE in the form $a\,p + b\,q = c$.
- Step 2: Write the auxiliary equations $\dfrac{dx}{a} = \dfrac{dy}{b} = \dfrac{du}{c}$.
- Step 3: Find one first integral $\varphi(x, y, u) = c_1$.
- Step 4: Find a second, independent first integral $\psi(x, y, u) = c_2$.
- Step 5: Write the general solution $G(\varphi, \psi) = 0$.
- Two standard techniques for steps 3–4:
- (a) the method of grouping, (b) the method of multipliers.
- A refinement of (b): the equivalent ratio (eigenvalue) method.



Method of Grouping

- Consider the auxiliary equations

$$\frac{dx}{a(x,y,u)} = \frac{dy}{b(x,y,u)} = \frac{du}{c(x,y,u)}. \tag{1}$$

- Take any two of the three ratios in (1) and equate them.
- Cancel common factors from the denominators, if any.
- If the result involves only two variables, it is an ODE.
- Integrate it to get $\varphi(x,y,u) = c_1$.
- Take another pair of ratios and repeat to get $\psi(x,y,u) = c_2$.
- The two together give the general solution $G(\varphi,\psi) = 0$.



Grouping with Substitution

- Often a pair of ratios still contains a third variable.
- Then use the first integral $\varphi(x, y, u) = c_1$ to express that variable in terms of the other two.
- Example: from $x - y = c_1$, replace x by $y + c_1$.
- Integrate the resulting ODE treating c_1 as a constant.
- This gives $\psi(x, y, u; c_1) = c_2$.
- Finally replace c_1 by $\varphi(x, y, u)$.
- Result: second first integral $\psi(x, y, u) = c_2$.
- Do not forget the final back-substitution.



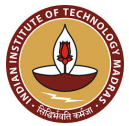
Example: $u_x + u_y = 1$

- Here $a = 1,\ b = 1,\ c = 1$.
- Auxiliary equations: $\dfrac{dx}{1} = \dfrac{dy}{1} = \dfrac{du}{1}$.
- $dx = dy \Rightarrow x - y = c_1$.
- $dy = du \Rightarrow u - y = c_2$.
- General solution: $G(x - y,\ u - y) = 0$.
- Equivalently $u = y + g(x - y)$, g arbitrary.
- Check: $u_x = g'$, $u_y = 1 - g'$, so $u_x + u_y = 1$ ✓.



Example: $x\,p + y\,q = u$

- Auxiliary equations: $\dfrac{dx}{x} = \dfrac{dy}{y} = \dfrac{du}{u}$.
- First pair: $\ln y = \ln x + \ln c_1 \Rightarrow \dfrac{y}{x} = c_1$.
- First and third: $\ln u = \ln x + \ln c_2 \Rightarrow \dfrac{u}{x} = c_2$.
- General solution: $G\left(\dfrac{y}{x}, \dfrac{u}{x}\right) = 0$.
- Equivalently $\dfrac{u}{x} = g\left(\dfrac{y}{x}\right)$, i.e. $u = x\,g(y/x)$.
- These are exactly the functions homogeneous of degree one.
- Compare with the formation example $u = x^n f(y/x)$ with $n = 1$.



Example: $x^2p + y^2q = (x+y)\,u$

- Auxiliary equations: $\dfrac{dx}{x^2} = \dfrac{dy}{y^2} = \dfrac{du}{(x+y)u}$.
- First pair: $\dfrac{dx}{x^2} = \dfrac{dy}{y^2} \Rightarrow \dfrac{1}{x} - \dfrac{1}{y} = c_1$.
- The other pairs contain three variables; combine ratios instead.
- Subtract numerators and denominators of the first two:

$$\frac{dx - dy}{x^2 - y^2} = \frac{du}{(x+y)u} \Rightarrow \frac{d(x-y)}{x-y} = \frac{du}{u}.$$

- Integrating: $\dfrac{x-y}{u} = c_2$.
- General solution: $G\left(\dfrac{1}{x} - \dfrac{1}{y},\ \dfrac{x-y}{u}\right) = 0$.
- Since $\frac{y-x}{xy} \cdot \frac{u}{x-y} = -\frac{u}{xy}$, also $G\left(\frac{xy}{u}, \frac{x-y}{u}\right) = 0$.



Example: $(x+y)\,p+(x+y)\,q=-(x+y+2u)$

- Auxiliary equations: $\dfrac{dx}{x+y}=\dfrac{dy}{x+y}=\dfrac{du}{-(x+y+2u)}$.
- First pair: $dx=dy\Rightarrow x-y=c_1$.
- Second and third, replacing $x=y+c_1$:

$$\frac{dy}{2y+c_1}=\frac{du}{-(2y+2u+c_1)}.$$

- $\Rightarrow (2y+2u+c_1)\,dy+(2y+c_1)\,du=0.$
- $\Rightarrow 2y\,dy+c_1\,dy+c_1\,du+2(u\,dy+y\,du)=0.$
- Exact: $d\left(y^2+c_1y+c_1u+2uy\right)=0.$

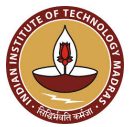


Example (contd.)

- Integrating: $y^2 + c_1 y + c_1 u + 2uy = c_2$.
- Replace $c_1 = x - y$:
- $y^2 + (x - y)y + (x - y)u + 2uy = c_2$.
- Simplify: $xy + yu + ux = c_2$.
- General solution:

$$G\big(x - y,\ xy + yu + ux\big) = 0.$$

- Check: multipliers $(y + u,\ x + u,\ x + y)$ give $(x + y)\big[(y + u) + (x + u) - (x + y + 2u)\big] = 0$ ✓.



Example: $x(y^2+u)\,p - y(x^2+u)\,q = (x^2-y^2)\,u$

- Auxiliary equations: $\dfrac{dx}{x(y^2+u)} = \dfrac{dy}{-y(x^2+u)} = \dfrac{du}{(x^2-y^2)u}$.
- Try multipliers $\frac{1}{x}$, $\frac{1}{y}$, $\frac{1}{u}$: the combined denominator is

$$(y^2+u) - (x^2+u) + (x^2-y^2) = 0.$$

- Hence the numerator vanishes: $\dfrac{dx}{x} + \dfrac{dy}{y} + \dfrac{du}{u} = 0$.
- $d(\ln xyu) = 0 \Rightarrow xyu = c_1$, i.e. $u = \dfrac{c_1}{xy}$.
- For the second integral, use $u = c_1/(xy)$ in the first two ratios.
- (Strategy: eliminate u and get an ODE in x and y only).



Example (contd.)

- $\dfrac{dx}{x\left(y^2+\frac{c_1}{xy}\right)} = \dfrac{dy}{-y\left(x^2+\frac{c_1}{xy}\right)}.$
- $\Rightarrow y\left(x^2+\dfrac{c_1}{xy}\right)dx + x\left(y^2+\dfrac{c_1}{xy}\right)dy = 0.$
- $\Rightarrow xy\,(x\,dx + y\,dy) + c_1\,\dfrac{y\,dx + x\,dy}{xy} = 0.$
- Divide by xy: $x\,dx + y\,dy + c_1\,\dfrac{d(xy)}{(xy)^2} = 0.$
- Integrate: $\dfrac{x^2}{2} + \dfrac{y^2}{2} - \dfrac{c_1}{xy} = c_2.$
- Replace $c_1/(xy) = u$: $x^2 + y^2 - 2u = c_3.$
- General solution: $G\left(xyu,\ x^2+y^2-2u\right) = 0.$

